

**BARNWELL, SC, USA**WMO: **720602**Lat: **33.250N**Lon: **81.383W**Elev: **75**StdP: **100.42**Time Zone: **-5.00 (NAE)**Period: **09-19**WBAN: **00194****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |              |            |            | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|--------------|------------|------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%          |            |            | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP           | HR         | MCDB       | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-4.0 | (c)<br>-2.2 | (d)<br>-12.9                  | (e)<br>1.2 | (f)<br>0.5 | (g)<br>-11.0 | (h)<br>1.5 | (i)<br>2.9 | (j)<br>8.5            | (k)<br>14.2 | (l)<br>7.6 | (m)<br>13.2 | (n)<br>0.9            | (o)<br>260 | (p)<br>0.368 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 36.2            | 24.6 | 35.0 | 24.7 | 33.7 | 24.2 | 26.7                | 32.4 | 26.1 | 31.8 | 25.7 | 31.1 | 3.0                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.2                            | 20.5 | 28.1 | 24.9 | 20.1 | 27.9 | 24.1 | 19.1 | 27.4 | 83.6          | 32.3 | 81.1 | 31.5 | 78.9 | 31.5 | 29.7              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.4               | 6.2  | 5.4 | -7.6                       | 38.0 | 2.1                | 1.6 | -9.1                                               | 39.1 | -10.3      | 40.0 | -11.5      | 40.9 | -13.0      | 42.1 |
|                   |      |     | -8.4                       | 27.4 | 2.1                | 0.5 | -9.9                                               | 27.8 | -11.2      | 28.1 | -12.3      | 28.4 | -13.8      | 28.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 18.7   | 8.2   | 10.9  | 14.0  | 18.4  | 23.1  | 26.8  | 28.0  | 27.2  | 24.8  | 19.0  | 12.6  | 10.4  |  |  |
|                                                                        | DBStd       | 8.03   | 5.54  | 5.40  | 5.19  | 3.94  | 3.15  | 2.22  | 1.78  | 1.85  | 2.67  | 4.24  | 4.38  | 5.17  |  |  |
|                                                                        | HDD10.0     | 255    | 102   | 51    | 22    | 1     | 0     | 0     | 0     | 0     | 1     | 20    | 59    |       |  |  |
|                                                                        | HDD18.3     | 1212   | 316   | 216   | 153   | 47    | 6     | 0     | 0     | 0     | 1     | 44    | 179   | 251   |  |  |
|                                                                        | CDD10.0     | 3412   | 47    | 74    | 145   | 254   | 406   | 504   | 558   | 532   | 444   | 278   | 98    | 72    |  |  |
|                                                                        | CDD18.3     | 1328   | 2     | 7     | 18    | 50    | 153   | 254   | 299   | 273   | 195   | 63    | 7     | 6     |  |  |
|                                                                        | CDH23.3     | 13132  | 11    | 47    | 183   | 567   | 1480  | 2671  | 3187  | 2624  | 1731  | 566   | 49    | 16    |  |  |
|                                                                        | CDH26.7     | 5876   | 1     | 3     | 36    | 167   | 617   | 1330  | 1582  | 1223  | 752   | 161   | 6     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.9    | 2.0   | 2.2   | 2.3   | 2.3   | 2.1   | 1.8   | 1.8   | 1.6   | 1.8   | 1.8   | 1.8   | 1.8   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1194   | 104   | 93    | 106   | 70    | 76    | 139   | 132   | 140   | 96    | 89    | 65    | 83    |  |  |
|                                                                        | PrecMax     | 1552   | 215   | 189   | 266   | 175   | 176   | 238   | 250   | 329   | 231   | 335   | 198   | 206   |  |  |
|                                                                        | PrecMin     | 954    | 42    | 32    | 17    | 23    | 24    | 46    | 71    | 74    | 18    | 0     | 9     | 13    |  |  |
|                                                                        | PrecStd     | 187    | 49    | 40    | 55    | 43    | 39    | 53    | 43    | 63    | 49    | 79    | 46    | 49    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.0  | 27.3  | 29.1  | 32.3  | 36.2  | 37.4  | 37.7  | 37.2  | 35.2  | 32.6  | 27.5  | 25.2  |  |  |
|                                                                        |             | MCWB   | 18.5  | 19.3  | 18.1  | 20.9  | 21.8  | 25.3  | 25.2  | 25.1  | 23.5  | 24.2  | 20.5  | 21.3  |  |  |
|                                                                        | 2%          | DB     | 22.3  | 24.8  | 27.6  | 30.2  | 33.0  | 36.1  | 36.2  | 35.0  | 33.0  | 30.1  | 24.1  | 22.8  |  |  |
|                                                                        |             | MCWB   | 17.1  | 18.0  | 18.3  | 19.7  | 21.7  | 24.6  | 25.0  | 25.1  | 23.4  | 22.4  | 17.8  | 19.3  |  |  |
|                                                                        | 5%          | DB     | 19.9  | 22.6  | 25.9  | 28.7  | 32.2  | 34.0  | 34.9  | 33.7  | 32.5  | 28.0  | 22.6  | 21.2  |  |  |
|                                                                        |             | MCWB   | 16.1  | 16.5  | 17.3  | 18.7  | 21.5  | 24.0  | 25.2  | 24.5  | 23.2  | 21.1  | 17.2  | 18.5  |  |  |
|                                                                        | 10%         | DB     | 17.7  | 20.2  | 22.9  | 27.2  | 30.2  | 32.8  | 33.6  | 32.5  | 31.3  | 27.0  | 21.1  | 18.9  |  |  |
|                                                                        |             | MCWB   | 14.4  | 15.9  | 15.7  | 18.5  | 21.5  | 23.9  | 25.1  | 24.5  | 23.1  | 20.3  | 17.2  | 17.1  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.1  | 21.0  | 21.0  | 23.7  | 24.9  | 27.6  | 27.4  | 27.2  | 26.1  | 25.9  | 22.4  | 23.1  |  |  |
|                                                                        |             | MCDB   | 22.0  | 24.1  | 25.7  | 28.5  | 29.6  | 32.4  | 33.5  | 32.7  | 31.5  | 29.0  | 24.4  | 23.8  |  |  |
|                                                                        | 2%          | WB     | 18.6  | 19.7  | 20.0  | 22.1  | 23.8  | 26.3  | 26.7  | 26.4  | 25.3  | 24.7  | 21.0  | 21.1  |  |  |
|                                                                        |             | MCDB   | 21.7  | 23.4  | 24.5  | 26.5  | 28.5  | 31.9  | 33.2  | 31.6  | 30.3  | 27.6  | 22.7  | 21.9  |  |  |
|                                                                        | 5%          | WB     | 17.3  | 18.2  | 18.9  | 21.1  | 23.1  | 25.6  | 26.1  | 25.9  | 24.7  | 23.3  | 19.4  | 19.0  |  |  |
|                                                                        |             | MCDB   | 18.7  | 21.2  | 23.2  | 25.1  | 28.0  | 31.6  | 32.5  | 30.8  | 29.4  | 26.5  | 21.4  | 20.4  |  |  |
|                                                                        | 10%         | WB     | 15.4  | 16.8  | 17.6  | 20.0  | 22.6  | 25.0  | 25.6  | 25.5  | 24.1  | 22.3  | 17.7  | 17.3  |  |  |
|                                                                        |             | MCDB   | 17.3  | 19.4  | 21.0  | 24.2  | 27.4  | 31.0  | 31.5  | 30.2  | 28.4  | 25.2  | 19.9  | 18.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.0  | 12.6  | 13.4  | 13.5  | 12.2  | 11.6  | 10.9  | 10.4  | 11.4  | 12.9  | 13.0  | 11.2  |  |  |
|                                                                        |             | MCDBR  | 13.9  | 14.4  | 15.4  | 16.4  | 14.9  | 13.3  | 12.8  | 12.1  | 13.2  | 13.6  | 14.8  | 12.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 9.1   | 8.3   | 7.3   | 6.5   | 5.0   | 4.2   | 3.8   | 3.7   | 4.4   | 6.3   | 8.8   | 8.3   |  |  |
|                                                                        |             | MCWBR  | 11.0  | 11.8  | 12.4  | 11.4  | 10.9  | 11.8  | 11.2  | 10.3  | 10.7  | 10.2  | 10.9  | 9.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.325 | 0.333 | 0.368 | 0.404 | 0.470 | 0.509 | 0.551 | 0.529 | 0.437 | 0.401 | 0.348 | 0.322 |  |  |
|                                                                        | taud        |        | 2.488 | 2.463 | 2.395 | 2.287 | 2.143 | 2.082 | 1.986 | 2.039 | 2.283 | 2.345 | 2.462 | 2.536 |  |  |
|                                                                        | Ebn at Noon |        | 891   | 920   | 909   | 886   | 827   | 792   | 756   | 766   | 828   | 831   | 855   | 875   |  |  |
|                                                                        | Edh at Noon |        | 87    | 98    | 114   | 132   | 154   | 164   | 179   | 168   | 126   | 110   | 89    | 79    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.85  | 3.48  | 4.55  | 5.74  | 6.19  | 6.33  | 6.09  | 5.55  | 4.79  | 4.05  | 3.12  | 2.49  |  |  |
|                                                                        | RadStd      |        | 0.15  | 0.35  | 0.34  | 0.40  | 0.44  | 0.45  | 0.43  | 0.29  | 0.41  | 0.47  | 0.23  | 0.23  |  |  |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (51 neighbors) |  | +0.56 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +188    | +130    |

Nomenclature: See separate page